0. **Code:** karuturi m 2445.zip

1. final b / Program.cs
2. final c / Program.cs
3. final exam / Program.cs

1. **final b / Program.cs**

```
namespace final_b
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] array = {
                23,
                1,
                10,
                5,
                2
            };

            int[] result = insersion(array);

            Console.WriteLine(string.Join(" ,", result));
        }

        static int[] insersion(int[] array)
        {
            int len = array.Length;

            int[] count = new int[len];

            for (int i = 0; i < len; i++)
            {
                int key = array[i];
                int j = i - 1;

                while (j ≥ 0 && array[j] > key)
```

B1

B2

B3

1. **final b / Program.cs**

```
                {
                    array[j + 1] = array[j];
                    j--;
                    count[i]++;
                    array[j + 1] = key;


                }

            }
            Console.WriteLine(String.Join(",", array));
            return count;


        }


    }


}
```

array[j+1] = key Need to

needs to insert at the end.

B3

or it won't work.

B4

Sorting an array

2. final c / Program.cs

```
namespace final_c
{
    internal class Program
    {
        static void Main(string[] args)
        {
            int[] array = {
                1,
                2,
                3,
                2,
                7,
                3,
                4,
                4,
                5,
                5,
                4,
                5,
                1,
                6,
                1,
                1
            };
            int result = largenum(array);

            Console.WriteLine("sum for max freq " + result);
        }

        static public int largenum(int[] array)
        {
```

2. **final c / Program.cs**

```
            Dictionary < int, int > frequency = new Dictionary < int, int > ();
            Dictionary < int, int > maxdict = new Dictionary < int, int > ();
            int sum = 0;

            foreach(int i in array)
            {
                if (frequency.ContainsKey(i))
                {
                    frequency[i]++;

                }

                else
                {
                    frequency[i] = 1;


                }


            }
            foreach(var kvp in frequency)
            {
                int key = kvp.Key;
                int value = kvp.Value;


                if (!maxdict.ContainsKey(value) || key > maxdict[value])
                {
```

if(array==null) return 0;

C2

num → Count
key Value

(Value)
Count →
max num
(key)

C3

2. final c / Program.cs

```
                    maxdict[value] = key;

                }

            }

            foreach(var kvp2 in maxdict)
            {
                sum = sum + kvp2.Value * kvp2.Key;


            }

            return sum;


        }
    }
}
```

good.

3. final exam / Program.cs

```
using System.ComponentModel;
using System.Diagnostics.CodeAnalysis;
using System.Security.Cryptography.X509Certificates;

namespace final_exam
{
    internal class Program
    {
        static void Main(string[] args)
        {
            Console.WriteLine("enter a number either positive or negative ");
            int integer = int.Parse(Console.ReadLine());



            int base_n;

            do {
                Console.WriteLine("enter a base value must be a positive number ");
                base_n = int.Parse(Console.ReadLine());

            } while (base_n < 0);

            int value = largest(integer, base_n);



        }
```

A1

A2

3. **final exam / Program.cs**

```
30
31      static public int largest(int interger, int base_n)
32      {
33
34          int max = int.MinValue;
35          int divnum = 0;
36          int sum = 0;
37
38          while (interger < 0)
39          {
40
41              interger = -interger;
42
43          }
44
45          while (interger > 0)
46          {
47              divnum = interger % base_n;
48              sum = sum + divnum;
49
50
51              if (divnum > max)
52              {
53                  max = divnum;
54
55              }
56
57              interger = interger / base_n;
58
59
60          }
```

0; → When interger = 0, max will be min Value? It should be 0.

A5

If

A3

A3

A4

A5

3. **final exam / Program.cs**

```
            Console.WriteLine($"max value is {max}");
            Console.WriteLine($"sum of the digits is {sum}");
            return sum;

        }
    }
}
```

A4

A5

Not Used